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Quantitative Measurement of a Chiral Drug in a Complex Matrix: A *J*-Compensated Quantitative HSQC NMR Method

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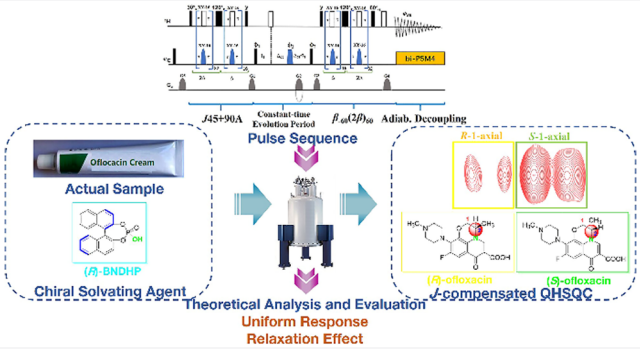
ABSTRACT: We developed a *J*-compensated QHSQC NMR method for the quantitative measurement of enantiopurity and concentration of levofloxacin in a complex drug matrix. The pulse sequence can achieve uniform signal responses by the suppression of the heteronuclear coupling modulation and alleviation of the homonuclear coupling evolution. Furthermore, we discuss the influence of relaxation on peak intensities and propose the selection conditions of reference signals to achieve accurate quantifications. The evaluation of the methodology shows that the results obtained by selected peaks are in accordance with theoretical analysis with good reliability and linearity. The enantiomeric separation and quantification of levofloxacin in creams are achieved by using a chiral solvating agent, a reference compound, and the *J*-compensated QHSQC pulse sequence. To our knowledge, this is the first example of QHSQC methodology to quantify enantiomers in the complex matrix.



Identification and determination of chirality of organic compounds in a complex matrix is still a fundamental and challenging task in the fields of organic chemistry and pharmaceutical analysis. Traditional methods for the quantitative analysis of enantiomers include chiral gas chromatography,[1,2] liquid chromatography,[3,4] supercritical fluid chromatography,[5−7] electrophoresis,[8−10] fluorescence spectroscopy,[11,12] and circular dichroism.[13−15] However, these all require laborious extraction and tedious separation steps. To improve separation procedures, recent research mainly focused on the development of novel materials to offer more cost-effective and quick solutions.[16−19] NMR spectroscopy with enantiopure chiral solvating agents has an enormous potential for the differentiation of enantiomers.[20−23] Enantiomeric discrimination can be efficiently allowed by mixing enantiomers and chiral solvating agents (CASs) due to the weak, noncovalent interactions between the analyte of interest and CASs. In most cases, the scopes of application are only limited to enantiomeric mixtures[24−28] or noncomplex samples.[29,30] In addition, the traditional applications of this method have always focused on screening different types or various functional groups of chiral solvating agents to improve unsatisfactory separation performance.[31−34]

The limited resolution of ^{1}H NMR spectra arising from the narrow spectral width and the crowded signals prevents these wider applications for the determination of enantiopurity, especially in complex matrixes. On the contrary, HSQC spectra have higher resolution than ^{1}H NMR spectra because signals are further spread into two dimensions.[35−37] Signal dispersion is high in the ^{13}C dimension which is characterized by a wide range of chemical shifts, and thus, HSQC experiments are a promising tool for assigning signals in overcrowded spectra of complex organic mixtures. Therefore, we hypothesize that the quantitative HSQC (QHSQC) methodology can enable an increase of the applications in the field of enantiomeric separation and determination.

However, peak volumes in the HSQC spectra acquired with the traditional pulse sequences (Figure 1A) are subject to the transfer period of the insensitive nuclei enhanced by polarisation transfer (INEPT) and retro-INEPT blocks, which could not simultaneously match a large range of different heteronuclear coupling constants ($^{1}J_{CH}$) values.[35] Because signal intensity is a function of $\sin^2(2\pi^1 J_{CH}\Delta)$, there are significant errors in quantifications which affect a method's precision. Moreover, different homonuclear couplings diversely evolve and finally modulate observable heteronuclear single quantum coherences during the whole pulse sequence, leading to undesirable amplitude attenuations and mixed-phase distortion.[38−40] Therefore, the effective compensation of

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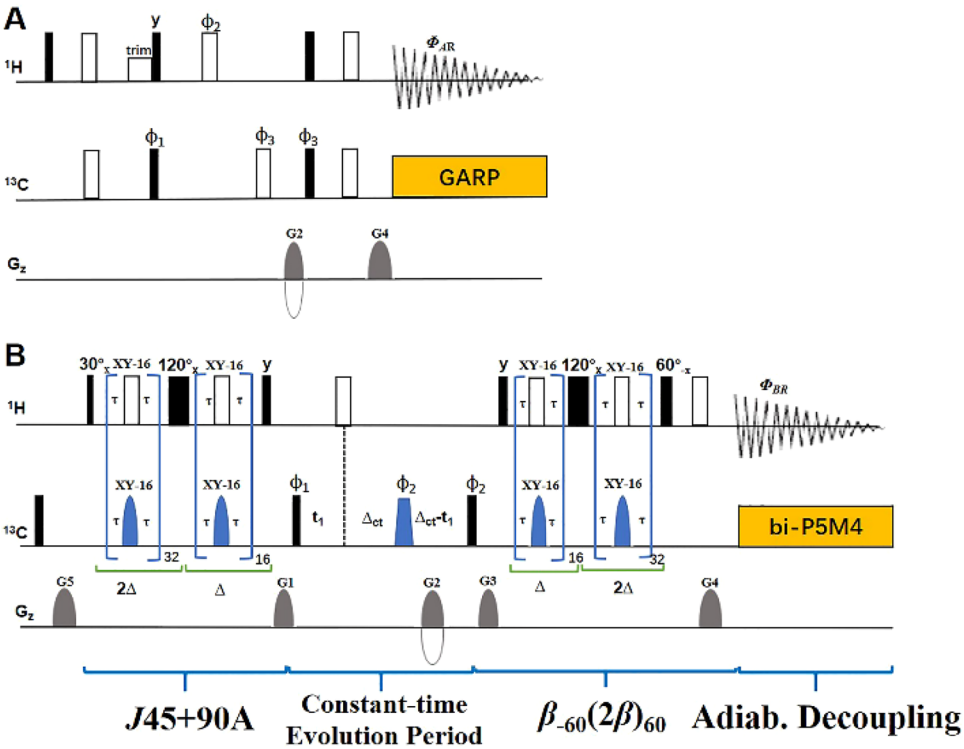

Figure 1. Pulse sequences of the traditional HSQC (A) and *J*-compensated QHSQC (B). Black filled and white open rectangles indicate hard 90° and 180° pulses, respectively, unless indicated otherwise. Blue sine bell shapes represent TanhTan inversion pulses. Blue trapezium shape represents chirp refocusing pulses (Crp60com.4). Gradients used for coherence selection are set to G2 = ± 80.0% and G4 = 20.1% of the maximum gradient strength of 57.7 G/cm. Gradients used as homospoils are set to G1 = 23.0%, G3 = 33.0%, and G5 = ± 30.0%. The 2Δ and Δ delays (Δ = 3.45 ms) are accomplished by simultaneously introducing 32 and 16 pulses with XY-16 cycling simultaneously on proton and carbon channel.[38] The τ-delay in CPMG-INEPT is set to 82.2 μs, which was short enough to suppress the J_{HH} coupling evolution. Constant-time evolution period (Δ_{ct} = 18.5 ms) is designed to reduce J_{CC} coupling modulation.[54,55] The pulse phases are along *x* if not indicated otherwise. Phase cycles are $\phi_1 = x, -x$; $\phi_2 = 2(x), 2(-x)$; $\phi_3 = 4(x), 4(-x)$; $\phi_{AR} = 2(x, -x), 2(-x, x)$; $\phi_{BR} = x, -x, -x, x$. The XY-16 is applied with phases [$x, y, x, y, y, x, y, x, -x, -y, -x, -y, -y, -x, -y, -x$]. The axial peak suppression is accomplished with the echo-antiecho method by inverting the phase XY-16 on carbon channel during the INEPT period, ϕ_1 and ϕ_R on every second increment of evolution time. The adiabatic decoupling and reduction of undesired cycling sidebands are achieved with Crp(32, 1.5, 20.2) using the M4P5 supercycle on the ^{13}C channel.

heteronuclear coupling modulations and the suppression of homonuclear evolutions are the keys for accurate quantifications. The original QHSQC methods relied on selecting four polarization transfer values with different delays during the constant-time INEPT step.[41,42] These pulse sequences and their modifications produced relatively uniform intensity response over a $^1J_{CH}$ range of 115−190 Hz by averaging all signals.[41,42] Another recently published approach was the HSQC$_0$ sequence, which was based on the extrapolation of peak volumes by recording a series of HSQC spectra with successive repetitions of the same pulse train.[43,44] This method ingeniously evaded the inherent flaw of the pulse sequence and confirmed that the signal attenuation was proportional to the repetition number. In addition, the existing and very efficient techniques for the *J*-compensated versions of the HSQC pulse sequence have also been presented, most notably the COB-HSQC,[45] the *J*-compensated HSQC by a pair of opposed adiabatic pulses,[46] and a plethora of related publications.[47,48] Despite efforts, current sequences still suffer from phase distortion, response nonuniformity, and sensitivity loss caused by homonuclear and heteronuclear coupling evolution. Herein, we present a QHSQC sequence that provides a rapid and reliable quantification tool. The new sequence was successfully combined with a chiral solvating agent for the determination of the enantiopurity of levofloxacin.

■ EXPERIMENTAL SECTION

Chemicals. Strychnine (99+%), levofloxacin (98+%), ofloxacin (99+%), $CDCl_3$, and DMF-d_6 were purchased from J&K Scientific Ltd. The 5 mm NMR tubes were purchased from Norell. The actual cream matrix for ofloxacin cream was purchased from Zhejiang Pharmaceutical Technology Company.

Quantitative NMR Measurement. The *J*-compensated QHSQC experiments were performed on a 400 MHz Bruker AVANCE III spectrometer equipped with a 5 mm BBFO probe. Data points (2048 × 512) with spectral width 16 and 210 ppm, respectively, with 16 scans per FID and interscan delay D_1 of 5.0 s (longer than 5 times of the longest selected peaks' T_1 values), resulting in a total acquisition time of 12.0 h for each HSQC. The Topspin 2.1 Bruker NMR software was applied for data processing.

Quantitative Calculation. The quantitative calculation of the absolute concentration of levofloxacin in complex matrix was followed:

$$C_s = \frac{V_s}{V_{ref}} \times \frac{N_{ref}}{N_s} \times C_{ref} \times P_{ref} \quad (4)$$

where V_s, N_s, and C_s are the peak volume, the number of the proton of the quantitative peak, and the absolute concentration of levofloxacin, respectively; V_{ref}, N_{ref}, C_{ref}, and P_{ref} are the peak

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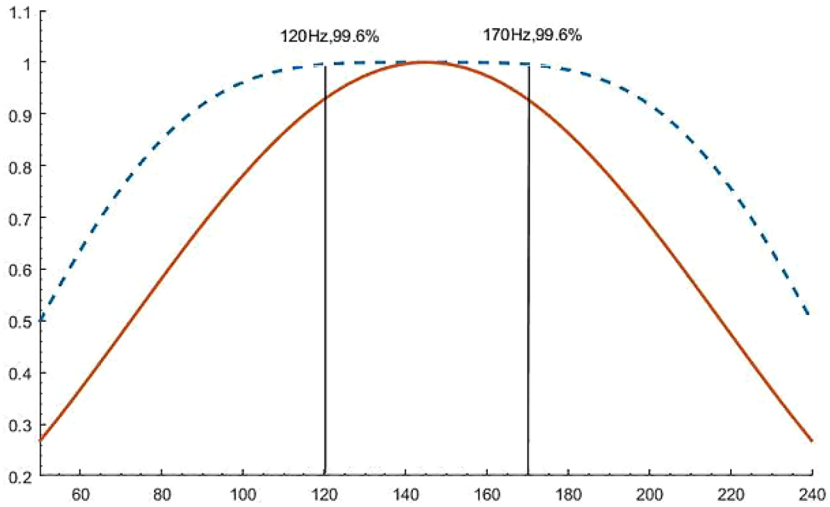

Figure 2. Theoretical peak volumes of traditional HSQC (solid red line) and *J*-compensated QHSQC (dashed blue line). For traditional HSQC, the signal performance depends on $\sin^2(2\pi^1J_{CH}\Delta)$ with Δ = 1.74 ms ($^1J_{CH}$ = 145 Hz). For *J*-compensated QHSQC, the signal response profile is followed by eq 1 with Δ = 3.45 ms ($^1J_{CH}$ = 145 Hz).

volume, the number of protons of the quantitative peak, the gravimetric concentration, and purity of the selected reference peak, respectively.

■ RESULTS AND DISCUSSION

A crucial factor that determines the quantitative reliability of a QHSQC sequence is the sophisticated manipulation of heteronuclear and homonuclear coupling evolutions. To simultaneously compensate $^1J_{CH}$ modulation and suppress J_{HH} evolution during INEPT and retro-INEPT blocks, the *J*-compensated Levitt's $\beta_x(2\beta)_{x+120}$ (or $\beta_{-60}(2\beta)_{60}$)[49] and its variant-*J*45 + 90A[50] with Carr−Purcell−Meiboom−Gill (CPMG) pulse train[51] are combined together. Additionally, to eliminate or lessen variations in B_1-inhomogeneity and the off-resonance effect, hard 180° inversion and refocusing pulses on ^{13}C are replaced by adiabatic broadband inversion (TanhTan)[52] and refocusing pulses (Crp60com.4). Meanwhile, the constant-time method is applied to suppress J_{CC} modulations and noise ridge, resulting in decreasing line shape broadening of the indirect dimension.[53,54] Moreover, adiabatic pulses are introduced to improve effective decoupling bandwidth in the ^{13}C channel, as well as minimize power and sampling heating.[55] Based on the above characteristics, this pulse sequence (Figure 1B) is referred to as the *J*-compensated QHSQC.

The modulation of $^1J_{CH}$ values on the response profile of the QHSQC experiment was examined and compared with the traditional HSQC. In the traditional HSQC, peak volumes of different functional groups depend upon the different spins signal responses, which are modulated as a function $\sin^2(2\pi^1J_{CH}\Delta)$.[41,42,47,48] Therefore, polarization transfer efficiency is dominated by an optimal delay time Δ and a selected $^1J_{CH}$ value. In complex mixtures the ingredients of interest coexist with larger compounds that cause the line shape broadening, which make the accurate measurement of coupling constants difficult or impossible. As a result, the mismatch between the optimal coupling constant and the true coupling constant leads to decrease in sensitivity and accuracy. In contrast, the new sequence gains much more uniform signal response than the original QHSQC. The intensity of the observable signal (I) prior to the acquisition period in the *J*-compensated pulse sequence is given by eq 1:

$$I \propto [0.75\sin(\pi^1J_{CH}\Delta) - 0.25\cos(2\pi^1J_{CH}\Delta)\sin(\pi^1J_{CH}\Delta) + 0.5\cos(\pi^1J_{CH}\Delta)\sin(2\pi^1J_{CH}\Delta)]^2 \quad (1)$$

A detailed analysis of *J*-compensated QHSQC pulse sequence, using the product operator formalism with the help of POMA,[56,57] is given in the Supporting Information I. The comparison of theoretical signal performance between traditional HSQC and *J*-compensated QHSQC is illustrated in Figure 2. It shows that a uniform response curve of the new pulse sequence can be achieved in the large range of $^1J_{CH}$ values (120−170 Hz), which empirically covers most aliphatic, alkenyl, and aromatic coupling constants. The overall theoretical variation of signal responses is less than ±0.4% over a range of $^1J_{CH}$ (120−170 Hz) and less than ±0.1% over a range of $^1J_{CH}$ (127−162 Hz). Therefore, this *J*-compensated pulse sequence is an efficient method to make effective transfers of quantum signals without $^1J_{CH}$ modulations.

The homonuclear couplings that evolve during the whole pulse sequence lead to undesired strongly distorted phases of the signals. As a result, integration errors make peak volumes to significantly deviate from the true concentrations, especially when a proton has several coupling partners and large J_{HH} values. This latter problem can be circumvented by using the CPMG-INEPT on both proton and carbon channels to suppress J_{HH} evolution. In order to simultaneously satisfy the suppression condition ($\tau < 1/(2\Delta\nu_{max})$) and the broadband inversion for a large spectra region, the standard Bruker adiabatic pulse (TanhTan)[53] was found to be an excellent choice, when the duration of adiabatic pulse and the delay τ were optimized with 50.0 and 82.2 μs, respectively. The XY-16 phase cycling can decrease inhomogeneities of the 180° pulse and ultimately improve the quantitative accuracy of the method.[38] Although a majority of samples in pharmaceutical analysis are of natural ^{13}C abundance, ^{13}C-labeled sample are also analyzed in several circumstances. This pulse sequence is appropriate for this type of samples as it deals with $^{13}C-^{13}C$

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coupling.[54,55] The constant-time period was found to decrease the phase distortion and artifacts in the F_1 dimension.

The relaxation effect is a significant source of signal loss in long pulse sequences and adversely impacts the sensitivity and the reliability of quantification. It should be noted the *J*-compensated QHSQC pulse sequence incorporates more pulses and delays than the traditional HSQC sequences and thus is more susceptible to relaxation effects. The influence of relaxation on peak intensity can be roughly described by the function[41]

$$I \propto \exp(-t/T_2) \tag{2}$$

where t refers to the time during which the proton magnetization is located on transverse plane before acquisition and T_2 to the transverse relaxation time of the proton coupled to the carbon-13. When an internal standard is used to determine the concentration of an analyte with a relatively large T_2 difference from the reference peak, the accurate measurement is influenced by the relaxation ratio between the target and the reference signal. The influence of the difference in T_2 values is described by factor η, which is defined as

$$\eta = \exp(-t/T_{2\mathrm{ref}})/\exp(-t/T_2) \tag{3}$$

where the factor η depends on the transverse relaxation times of the target and the reference signal, T_2 and $T_{2\mathrm{ref}}$ respectively. For small molecules in nonviscous liquids, T_2 empirically varies between 0.30 and 2.0 s. In Figure 3, the relaxation effect on the pulse sequence is evaluated by five different $T_{2\mathrm{ref}}$ values with a range of T_2 (0.20−2.0 s). The result shows that η depends on $T_{2\mathrm{ref}}$, which should follow two conditions of the quantification peaks. Condition 1: the value range of $T_{2\mathrm{ref}}$ should be taken close to T_2 of the target signal as much as possible. For example, if $T_{2\mathrm{ref}}$ is 0.60 s to keep η within ±1.0%, the range of T_2 should be within 0.48−0.80 s (Figure 3). If $T_{2\mathrm{ref}}$ takes 0.60 s and η is kept within ±5.0%, an accurate quantification can be achieved with more than 0.27 s of T_2 (Figure 3). Condition 2: Signals with the longest possible T_2 values should be selected for the target analyte and the reference compound. In Figure 3, the larger the numbers that $T_{2\mathrm{ref}}$ values take, the larger the range of T_2 that can be used for quantitative peaks to keep η within ±1.0%. Moreover, when $T_{2\mathrm{ref}}$ and T_2 values are small, the sensitivity of the analysis is lower, as only a small amount of magnetization is detected. Generally, when the reference signal is selected with a proper $T_{2\mathrm{ref}}$, the relaxation effect does not significantly affect the accuracy of the measurement.

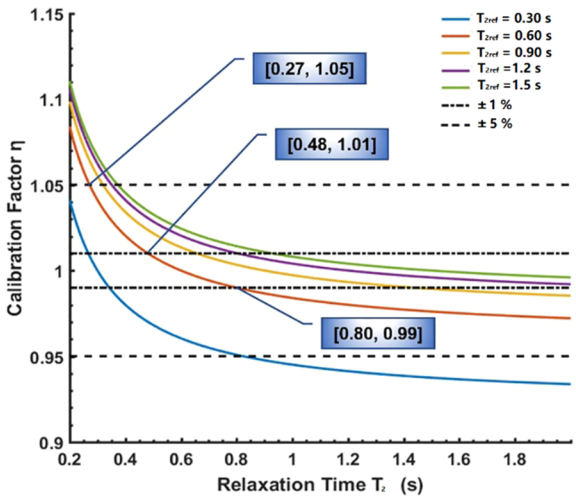

Figure 3. Factor (η) with respect to five different $T_{2\mathrm{ref}}$ values (0.30 (blue line), 0.60 (red line), 0.90 (yellow line), 1.2 (purple line), and 1.5 (green line) s) and a range of T_2 (0.20−2.0 s), which follows eq 3 with t = 0.0242 s. The dashed line and the dot-dashed line represent the ±5% and ±1% response factor of peak volumes, respectively.

The uniform response of the *J*-compensated QHSQC pulse sequence was demonstrated with a strychnine sample (Figure 4B), in contrast to the results obtained from the traditional HSQC spectrum. The peak volumes in the traditional HSQC experiment is more dependent on the coupling constants rather than the off-resonance effect and the relaxation effect (Table S1). The closer the coupling constants approach 145.0 Hz in the traditional HSQC experiments, the larger the peak volumes are acquired (Table S1). In comparison, the *J*-compensated QHSQC experiment yielded about 70.0 ppm flat response in the 40.0−100.0 ppm range of the F_1 dimension. The peak volumes outside of above the range declined sharply. One main reason for signal loss is the off-resonance effect caused by narrow inversion bandwidth of TanhTan pulse, when signals approaching the exciting edge, such as peak volumes of signals 1, 2, 3, 4, and 22. Another reason is the relaxation effect during the long pulse sequence before acquisition. If the T_2 of interest is shorter than 0.45 s and H11b is taken as a reference signal with T_2 (0.54 s), the loss of magnetization before acquisition becomes obvious. In addition, the baseline and phase properties of peaks is crucial to the success of quantification and should be investigated carefully. The phase distortion between two splits in one peak also prevents accurate quantification, as in the case of peak 8 in Figure 4A. These abnormalities probably were caused by phase dispersion and the antiphase coherence from a large amount of continuous 180° pulses, which arises from isotropic mixing of the coupled systems.[58] As compared to H11b as a reference signal, we found the standard deviation (SD) of relative peak volumes with the T_2 range of 0.45−0.60 s was below 6% (Figure 4B and Table S1). In comparison, the loss rate of peaks with T_2 below 0.45 s and above 0.60 s is too large and too small, respectively. As a result, the SD of the above two situations is over 6% (Table S1). Therefore, experimental results agree roughly with theoretical analysis of the relaxation effect. After comparing with the signal volumes of the traditional HSQC at the same transmitter offset, the signal response can keep on 93.3% of the maximum value in the *J*-compensated HSQC with the same scan number and similar acquisition time. As a result, the long pulse train and CPMG trains do not cause obvious signal loss. We evaluated the linearity of the peak areas against five concentrations (2.0, 4.0, 9.0, 15.0, and 20.0 g/L) of the strychnine sample in Figure 4C. The regression coefficient was 0.991, which indicates the potential of this quantitative method.

The capability of the new QHSQC sequence for the determination of enantiopurity in a complex mixture was evaluated using the actual levofloxacin drug. Levofloxacin or (*S*)-ofloxacin in Scheme 1 is a synthetic broad-spectrum antibacterial drug commonly used to treat a variety of bacterial infections.[59] Levofloxacin shows higher activity, increased water solubility, and fewer side effects than its (*R*)-enantiomer.[59] The (*R*)-enantiomer is possibly remained in the pill, gel, and cream due to impurities afforded by the synthetic process and the inappropriate purification process. It is therefore necessary to establish a powerful approach for the

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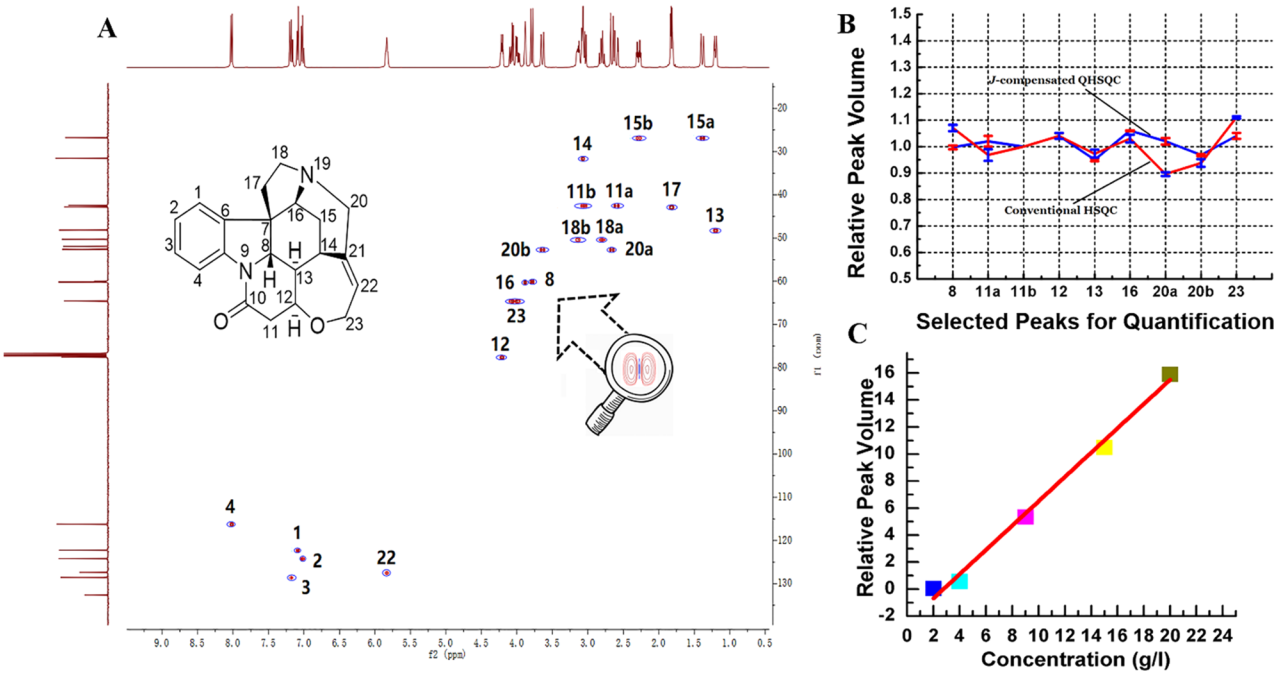
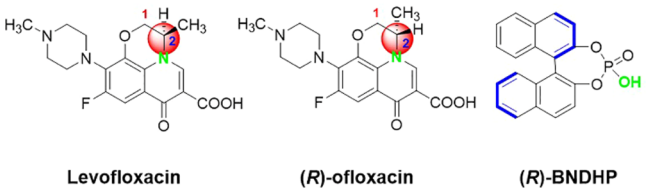


Figure 4. Evaluation of the *J*-compensated QHSQC pulse sequence. (A) The *J*-compensated QHSQC spectrum, integral areas, and assignments of strychnine sample (7.0 mg) in 0.5 mL of $CDCl_3$. (B) The relative peak volumes of the selected peaks for quantification by using 11b as reference with the *J*-compensated QHSQC and conventional HSQC. The error bar is the SD of relative peak volumes for three parallel experiments. (C) Linearity from the relative peak volumes of 11b against that of tetramethylsilane (TMS).

Scheme 1. Two Enantiomers of Ofloxacin and (*R*)-BNDHP



quantitative determination of the absolute concentration of the two enantiomers. This analysis is traditionally performed using high-performance liquid chromatography (HPLC). However, this approach involves multiple pretreatment steps including extraction, cleanup, and enrichment resulting in costing time, reagents, and samples. Moreover, HPLC methodologies require separating conditions optimization for different complex matrixes, while chiral columns are expensive. In contrast, NMR sample preparation is a relatively simple procedure that does not require recovery test and only includes sample weighing, spiking, and dissolution.

Enantiopurity of the topical ofloxacin can be determined using the *J*-compensated QHSQC pulse sequence. In the cream matrix, the signals around chiral center in ^{1}H NMR are overlapped by pharmaceutical excipients or influenced by protons of glyceryl monostearate and 1,2-propanediol, etc., as shown in Figure 5A. To ensure the clear peak separation in the HSQC spectra, several enantiopure chiral solvating agents and deuterated solvents were initially screened to have a great resolution. We found that (*R*)-1,1′-binaphthyl-2,2′-diyl hydrogen phosphate ((*R*)-BNDHP) (Scheme 1) can interact with ofloxacin to allow peak separation of enantiomers in the ^{1}H NMR spectra. It is assumed that the interaction is caused by the electrostatic attraction between the free hydroxyl group in (*R*)-BNDHP and the nitrogen atom (green marks) attached to the chiral center (red-ball marks) in ofloxacin. In this case, the protons at position 1 in ofloxacin are differently affected by the shielding effect of the binaphthyl group. As seen in Figure 5B, the axial protons of two enantiomers are clearly resolved. However, they are on the broad peak of the protons at oxygen atoms, and thus cannot be used for enantiopurity of quantitative analysis.

Surprisingly, it is found that the quantitative measurement of levofloxacin is not disturbed by other signals from the cream matrix, the active protons and the reference in QHSQC experiments, after comparing Figure 5C with 5D. The T_2 values of two enantiomeric protons are 0.310 and 0.309 s, respectively, which are close and thus are in accordance to Condition 1. In Figure 5D, two axial peak volumes of enantiomers at position 1 display 60.2% enantiomer excess (ee), in comparison with 60.3% ee from HPLC detection in Supporting Information III. The time required for the pretreatment of QHSQC experiments is only half an hour, much shorter than the 3 h required for the HPLC experiments. In terms of the absolute concentration of levofloxacin in cream, we used the cross peak of H12/C12 at 4.57/76.8 ppm in strychnine as a reference signal because of the proximity of the T_2 of H12 (0.301 s) and its chemical shift to two enantiomeric protons of ofloxacin in the complex matrix (Figure 5D). The calculated absolute concentration (182 mg/g) is in a very good agreement with the concentration determined by HPLC (183 mg/g).

■ CONCLUSION

In summary, the present *J*-compensated QHSQC pulse sequence can meet the requirements for quantification of chiral compounds in a complex matrix. The uniform signal responses are achieved by compensating heteronuclear coupling modulations and by suppressing the homonuclear

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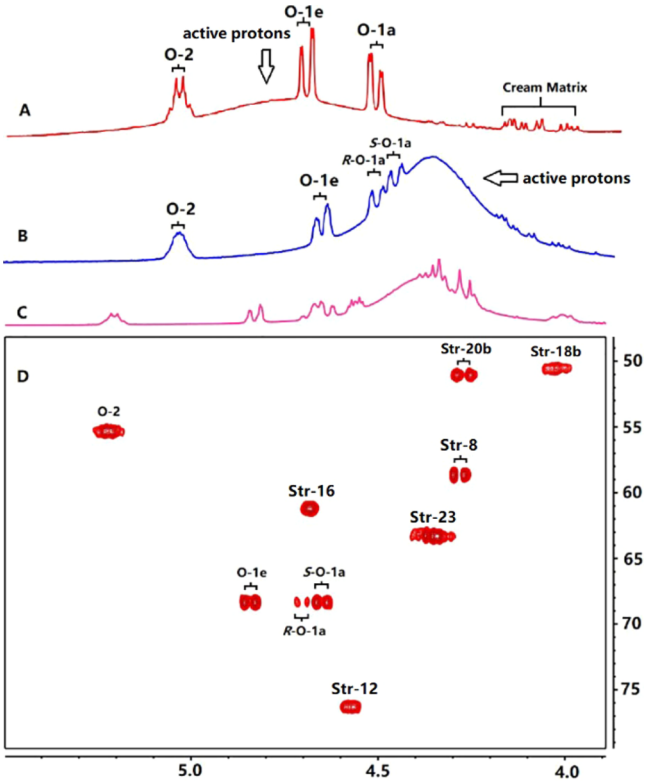

Figure 5. (A) The enlarged ^{1}H NMR spectrum of the actual ofloxacin cream (red line) in DMF-d_6. (B) The enlarged ^{1}H NMR spectrum of the actual ofloxacin cream and (*R*)-BNDHP (blue line) in DMF-d_6. (C) The enlarged ^{1}H NMR spectrum of the actual ofloxacin cream, (*R*)-BNDHP and strychnine (fuchsia line) in DMF-d_6. (D) The enlarged *J*-compensated QHSQC spectrum of the actual ofloxacin cream, (*R*)-BNDHP and strychnine (red contour map) in DMF-d_6. Notation: O (ofloxacin); Str (strychnine); *R* (*R* enantiomer); *S* (*S* enantiomer); a (the axial proton); e (the equatorial proton); numbers (assignments shown in Figure 4A and Scheme 1).

coupling evolutions during INEPT and retro-INEPT periods before acquisition. Based on the analysis of the relaxation effect, we sum up the two conditions for the selection of the most appropriate peaks to improve the reliability of quantification. With a simple pretreatment and the addition of (*R*)-BNDHP as a chiral solvating agent, the quantification of enantiopurity in the ofloxacin drug can be achieved successfully. In total, this study provides a unique route to utilize the two-dimensional quantitative NMR technique to resolve the stereochemistry in a complex matrix. The methodology has valuable prospects for the quantification of small organic compounds in complex biological and synthetic samples.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.9b04591.

Pulse sequence, theoretical analysis of *J*-compensated QHSQC pulse sequence by using the product operator formalism, raw NMR data, and the HPLC detection method (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Schurig, V. *J. Chromatogr. A* **2001**, *906*, 275−299.
(2) Zhang, J. H.; Xie, S. M.; Zhang, M.; Zi, M.; He, P. G.; Yuan, L. M. *Anal. Chem.* **2014**, *86*, 9595−9602.
(3) Castrignano, E.; Kannan, A. M.; Feil, E. J.; Kasprzyk-Hordern, B. *Chemosphere* **2018**, *206*, 376−386.
(4) Fiori, J.; Pasquini, B.; Caprini, C.; Orlandini, S.; Furlanetto, S.; Gotti, R. *J. Chromatogr. A* **2018**, *1562*, 115−122.
(5) Tao, Y.; Dong, F.; Xu, J.; Liu, X.; Cheng, Y.; Liu, N.; Chen, Z.; Zheng, Y. *J. Agric. Food Chem.* **2014**, *62*, 11457−11464.
(6) Schaffrath, M.; Weidmann, V.; Maison, W. *J. Chromatogr. A* **2014**, *1363*, 270−277.
(7) Tan, Q.; Fan, J.; Gao, R.; He, R.; Wang, T.; Zhang, Y.; Zhang, W. *Talanta* **2017**, *164*, 362−367.
(8) Patel, A. V.; Kawai, T.; Wang, L. P.; Rubakhin, S. S.; Sweedler, J. V. *Anal. Chem.* **2017**, *89*, 12375−12382.
(9) Yu, F. Z.; Zhao, Q.; Zhang, D. P.; Yuan, Z.; Wang, H. L. *Anal. Chem.* **2019**, *91*, 372−387.
(10) Zhang, Q. *TrAC, Trends Anal. Chem.* **2018**, *100*, 145−154.
(11) Mei, X. F.; Wolf, C. *Chem. Commun.* **2004**, 2078−2079.
(12) Tumambac, G. E.; Wolf, C. *Org. Lett.* **2005**, *7*, 4045−4048.
(13) Ding, K. L.; Ishii, A.; Mikami, K. *Angew. Chem., Int. Ed.* **1999**, *38*, 497−501.
(14) Nieto, S.; Dragna, J. M.; Anslyn, E. V. *Chem. - Eur. J.* **2010**, *16*, 227−232.
(15) Ghosn, M. W.; Wolf, C. *J. Am. Chem. Soc.* **2009**, *131*, 16360−16361.
(16) Banerjee-Ghosh, K.; Dor, O. B.; Tassinari, F.; Capua, E.; Yochelis, S.; Capua, A.; Yang, S.-H.; Parkin, S. S. P.; Sarkar, S.; Kronik, L.; Baczewski, L. T.; Naaman, R.; Paltiel, Y. *Science* **2018**, *360*, 1331−1334.
(17) Navarro-Sánchez, J.; Argente-García, A. I.; Moliner-Martínez, Y.; Roca-Sanjuán, D.; Antypov, D.; Campíns-Falcó, P.; Rosseinsky, M. J.; Martí-Gastaldo, C. *J. Am. Chem. Soc.* **2017**, *139*, 4294−4297.
(18) Wattanakit, C.; Côme, Y. B. S.; Lapeyre, V.; Bopp, P. A.; Heim, M.; Yadnum, S.; Nokbin, S.; Warakulwit, C.; Limtrakul, J.; Kuhn, A. *Nat. Commun.* **2014**, *5*, 3325.
(19) Dutta, S.; Gellman, A. J. *Chem. Soc. Rev.* **2017**, *46*, 7787−7839.
(20) Pirkle, W. H. *J. Am. Chem. Soc.* **1966**, *88*, 1837.
(21) Wenzel, T. J. In *Discrimination of Chiral Compounds Using NMR Spectroscopy*; Wiley Press: Hoboken, 2007.
(22) Parker, D. *Chem. Rev.* **1991**, *91*, 1441−1457.
(23) Uccello-Barretta, G.; Balzano, F. *Top. Curr. Chem.* **2013**, *341*, 69−131.
(24) Seo, M.-S.; Kim, H. *J. Am. Chem. Soc.* **2015**, *137*, 14190−14195.
(25) Monteagudo, E.; Virgili, A.; Parella, T.; Pérez-Trujillo, M. *Anal. Chem.* **2017**, *89*, 4939−4944.
(26) Khanvilkar, A. N.; Bedekar, A. V. *Chem. Commun.* **2018**, *54*, 11037−11040.
(27) Li, Y.; Yang, G.-H.; He, C. Q.; Li, X.; Houk, K. N.; Cheng, J.-P. *Org. Lett.* **2017**, *19*, 4191−4194.
(28) Li, G.; Cao, J.; Zong, W.; Lei, X.; Tan, R. *Org. Chem. Front.* **2016**, *3*, 96−102.
(29) Pérez-Trujillo, M.; Lindon, J. C.; Parella, T.; Keun, H. C.; Nicholson, J. K.; Athersuch, T. J. *Anal. Chem.* **2012**, *84*, 2868−2874.
(30) Molaabasi, F.; Talebpour, Z. *J. Agric. Food Chem.* **2011**, *59*, 803−808.
(31) Yang, L.; Wenzel, T.; Williamson, R. T.; Christensen, M.; Schafer, W.; Welch, C. J. *ACS Cent. Sci.* **2016**, *2*, 332−340.
(32) Huang, H.; Bian, G.; Zong, H.; Wang, Y.; Yang, S.; Yue, H.; Song, L.; Fan, H. *Org. Lett.* **2016**, *18*, 2524−2527.
(33) Du, G.; Li, Y.; Ma, S.; Wang, R.; Li, B.; Guo, F.; Zhu, W.; Li, Y. *J. Nat. Prod.* **2015**, *78*, 2968−2974.
(34) Feng, L.; Gao, G.; Zhao, H.; Zheng, L.; Wang, Y.; Stavropoulos, P.; Ai, L.; Zhang, J. *J. Org. Chem.* **2018**, *83*, 13874−13887.
(35) Giraudeau, P. *Magn. Reson. Chem.* **2014**, *52*, 259−272.
(36) Giraudeau, P. *Magn. Reson. Chem.* **2017**, *55*, 61−69.
(37) Koskela, H. *Annu. Rep. NMR Spectrosc.* **2009**, *66*, 1−31.
(38) Koskela, H.; Kilpeläinen, I.; Heikkinen, S. *J. Magn. Reson.* **2005**, *174*, 237−244.
(39) Luy, B.; Marino, J. P. *J. Am. Chem. Soc.* **2001**, *123*, 11306−11307.
(40) Castañar, L.; Sistaré, E.; Virgili, A.; Williamson, R. T.; Parella, T. *Magn. Reson. Chem.* **2015**, *53*, 115−119.
(41) Heikkinen, S.; Toikka, M. M.; Karhunen, P. T.; Kilpeläinen, I. A. *J. Am. Chem. Soc.* **2003**, *125*, 4362−4367.
(42) Mauve, C.; Khlifi, S.; Gilard, F.; Mouille, G.; Farjon, J. *Chem. Commun.* **2016**, *52*, 6142−6145.
(43) Hu, K.; Westler, W. M.; Markley, J. L. *J. Am. Chem. Soc.* **2011**, *133*, 1662−1665.
(44) Hu, K.; Ellinger, J. J.; Chylla, R. A.; Markley, J. L. *Anal. Chem.* **2011**, *83*, 9352−9360.
(45) Ehni, S.; Luy, B. *J. Magn. Reson.* **2014**, *247*, 111−117.
(46) Kupče, Ē.; Freeman, R. *J. Magn. Reson.* **1997**, *127*, 36−48.
(47) Mäkelä, V.; Helminen, J.; Kilpeläinen, I.; Heikkinen, S. *J. Magn. Reson.* **2016**, *271*, 34−39.
(48) Jiang, B.; Xiao, N.; Liu, H.; Zhou, Z.; Mao, X.; Liu, M. L. *Anal. Chem.* **2008**, *80*, 8293−8298.
(49) Levitt, M. H. *J. Magn. Reson.* **1982**, *48*, 234−264.
(50) Torres, A. M.; Bubb, W. A.; Philp, D. J.; Kuchel, P. W. *J. Magn. Reson.* **2008**, *194*, 81−88.
(51) Allerhand, A. *J. Chem. Phys.* **1966**, *44*, 1−9.
(52) Mulder, F. A. A.; de Graaf, R. A.; Kaptein, R.; Boelens, R. *J. Magn. Reson.* **1998**, *131*, 351−357.
(53) Hallenga, K.; Lippens, G. M. *J. Biomol. NMR* **1995**, *5*, 59−66.
(54) Vuister, G.; Bax, A. *J. Magn. Reson.* **1992**, *98*, 428−435.
(55) Claridge, T. D. C. In *High-Resolution NMR Techniques in Organic Chemistry*, 3rd ed; Elsevier Science: Oxford, 2016.
(56) Güntert, P. *Int. J. Quantum Chem.* **2006**, *106*, 344−350.
(57) Güntert, P.; Schaefer, N.; Otting, G.; Wüthrich, K. *J. Magn. Reson., Ser. A* **1993**, *101*, 103−105.
(58) Braunschweiler, L.; Ernst, R. R. *J. Magn. Reson.* **1983**, *53*, 521−528.
(59) Fish, D. N.; Chow, A. T. *Clin. Pharmacokinet.* **1997**, *32*, 101−119.

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